

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	SUMETHA SRI.S	Reg. No.:	192521097
Section / Batch:		Date:	24/07/2026

Code of Conduct

I SUMETHA SRI.S/192521097 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sumetha

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
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Date: _____	Date: _____	Date: _____

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PROFESSIONAL

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Array Search
Tracks

Q2
Stack
Tracks

Q3
Stack
Tracks

Q4
Stack
Tracks

Q5
Stack - Relating Symbols
Tracks

Q6
Memory Addressing Unit
Tracks

Q7
Stack
Tracks

Q8
Linked List
Tracks

Q9
Linked List
Tracks

Q10
Queue
Tracks

10/10 answered

Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<iostream.h>  
#define MAX 5  
int stack[MAX];  
int top=-1;  
int pop(){  
    if(top==0){  
        printf("Underflow\n");  
        return -1;  
    }  
    return stack[top--];  
}  
  
int main(){  
    int i;  
    stack[i]=10;  
    stack[i]=20;  
    stack[i]=30;  
    printf("Deleted Element: %d\n",pop());  
    printf("Deleted element: %d\n",pop());  
    printf("Deleted element: %d\n",pop());  
    return 0;  
}
```

Test Input

stdin input (optional)

Output

```
Deleted Element: 30  
Deleted element: 20  
Deleted Element: 10  
Underflow
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == '}') {  
    while(stack[top] != '}') {  
        postfix[j++] = pop0;  
    }  
    pop0;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(ch=='}') {  
2     while(top!=-1&& stack[top]!='') {  
3         {  
4             postfix[j++] = pop();  
5         }  
6         if(top!=-1&&7 stack[top]!='') {  
7             pop();  
8         }  
9     }
```

Test Input

stdin input (optional)

Output

Compilation Errors:
C:\Windows\TEMP\asandbox_1925210
97BA_66das96a23cf2.3094+801ma
in.c:111: error: expected
identifier or '(' before 'if'
111 if(ch=='}') {
112 while(top!=-1&& stack[top]!='') {
113 {
114 postfix[j++] = pop();
115 }
116 if(top!=-1&&7 stack[top]!='') {
117 pop();
118 }
119 }

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '[' && stack[top] == ']')) {
pop();
} else {
return 0;
}
}

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((ch=='&stack[top]=='[') ||  
2 (ch=='&stack[top]=='])')  
3 {  
4     pop();  
5 }  
6 else  
7 {  
8     return 0;  
9 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_1925210
97B3_6a6dae75aed125.37037380\ma
in.c:11: error: expected
identifier or ';' before 'if'
11: if(ch==']')
12: }

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10

Q6 Singly Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp=head;  
2 while (temp!=NULL)  
3 {  
4     printf("%d",temp->data);  
5     temp=temp->next;  
6 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_1925210
97RA_6a701c05377ee1.46262790\ma
in.c:11: warning: data
definition has no type or
storage class
11: temp=temp->next;
12: }

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

✓

Q2

Stack

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Stack - Balancing Symbols

5 marks

✓

Q6

Singly Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List

5 marks

✓

Q9

Linked List

5 marks

✓

Q10

Stack

5 marks

✓

Q7

Stack

5 marks

Point out the error.

```
POP(stack);
if top == 0;
print("Underflow")
else:
return stack[top--]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int POP(int stack[]){
2     if(top==0){
3         printf("Underflow");
4         return -1;
5     }
6     else{
7         return stack[top--];
8     }
9 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_1925210
97RA_6a701bdc3aa49.75173601\ma
in.c: In function 'POP':
C:\Windows\TEMP\sandbox_1925210
97RA_6a701bdc3aa49.75173601\ma
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

✓

Q2

Stack

5 marks

✓

Q3

Stack

5 marks

✓

Q4

Stack

5 marks

✓

Q5

Stack - Balancing Symbols

5 marks

✓

Q6

Singly Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List

5 marks

✓

Q9

Linked List

5 marks

✓

Q8

Linked List

5 marks

Identify the errors in the following code

```
while current != NULL:
current.next = prev
prev = current
current = current.next
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (current!=NULL){
2     next=current->next;
3     current->next=prev;
4     prev=current;
5     current=next;
6 }
7 head=prev;
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_1925210
97RA_6a6dafef719755.04911876\ma
in.c:11: error: expected
identifier or '(' before
while:
```

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q9 Linked List 5 marks

What is the error in the below code?
new.next = head
head = new

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new->next=head;
2 head=new;
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

Q10 Queue 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):
2 if(front==1){
3     print("Underflow");
4 }
5 else{
6     *Q[front++];
7     if(front>rear){
8         front=rear-1;
9     }
10    return *;
11 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_1928210
97RA_6e6daf2d8d6713.51772019\ma
in.c:11:1: warning: return type
defaults to 'int' [-Wimplicit-
int]

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QUESTIONS

Q1 Stack 5 marks ✓

Q2 Circular Queue 5 marks ✓

Q3 Stack 5 marks ✓

Q4 Stack 5 marks ✓

Q5 Doubly Linked List 5 marks ✓

Q6 Queue 5 marks ✓

Q7 Queue - Deque 5 marks ✓

Q8 Binary Search 5 marks ✓

Q9 Circular Queue 5 marks ✓

Q1 Stack 5 marks

Point out the error?
PEEK(stack);
return stack[top+1]

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int PEEK(int stack[])
2 {
3     if(top== -1) {
4         printf("Stack is Empty");
5         return -1;
6     }
7     return stack[top];
8 }
```

Test Input
stdin input (optional)

Output

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QUESTIONS

Q1 Stack 5 marks ✓

Q2 Circular Queue 5 marks ✓

Q3 Stack 5 marks ✓

Q4 Stack 5 marks ✓

Q5 Doubly Linked List 5 marks ✓

Q6 Queue 5 marks ✓

Q7 Queue - Deque 5 marks ✓

Q8 Binary Search 5 marks ✓

Q9 Circular Queue 5 marks ✓

Q2 Circular Queue 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
 printf("Queue Full");
}

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int vechile)
2 {
3     if((rear+1)%MAX==front){
4         printf("Queue Full");
5     }else{
6         rear=(rear+1)%MAX;
7         queue[rear]=vechile;
8     }
9 }
```

Test Input
stdin input (optional)

Output

Pending manual review

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q3 Stack 5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void PUSH(int item)
{
    if (top == MAX - 1) {
        printf("Overflow");
    }
    else {
        stack[++top] = item;
    }
}
```

Test Input
stdin input (optional)

Output

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q4 Stack 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if (top == -1)
{
    printf("Underflow");
}
```

Test Input
stdin input (optional)

Output

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Search

Windows Taskbar

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QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7 Queue - Deque 5 marks

Q8 Binary Search 5 marks

Q9 Circular Queue 5 marks

Q10 Stack - Priority Expression 5 marks

10/10 answered

Q7. Queue - Deque 5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void insertFront(int data) {  
2 // Check for overflow  
3 if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {  
4     return;  
5 }  
6  
7 // If deque is initially empty  
8 if (front == -1) {  
9     front = 0;  
10    rear = 0;  
11 }  
12 // If front is at the first position  
13 else if (front == 0) {  
14     front = MAX - 1;  
15 }  
16 // Decrement front standard way  
17 else {  
18     front = front - 1;  
19 }  
20 deque[front] = data;  
21 }  
22 }
```

Submitted

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\avndown_19312109
tmp_44767\sourcecode\10070275\main
in: In function 'insertFront':
C:\Windows\TEMP\avndown_19312109
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QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7 Queue - Deque 5 marks

Q8 Binary Search 5 marks

Q9 Circular Queue 5 marks

Q10 Stack - Priority Expression 5 marks

10/10 answered

Q8. Binary Search 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(l, mid, x);
return binarySearch(arr, mid, r, x);
}

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int l, int r, int x) {  
2 if (l > r) {  
3     return -1;  
4 }  
5 int mid = l + (r - l) / 2;  
6  
7 if (arr[mid] == x) {  
8     return mid;  
9 }  
10 if (arr[mid] > x) {  
11     return binarySearch(arr, l, mid - 1, x);  
12 }  
13 return binarySearch(arr, mid + 1, r, x);  
14 }
```

Submitted

Test Input

stdin input (optional)

Output

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